Forward Probability – Change of Numéraire

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Abstract

In this paper, we recall how the forward probability and change of numeraire framework can be applied to the pricing of various fixed income products.

Contents

1 Forward Probability – Change of Numeraire **1**
1.1 Money market contract 3

2 Pricing of a bond option **3**
2.1 Pricing an option on a zero-coupon bond 3
2.2 Pricing of a bond option at time 0 4

3 Pricing of a cap **4**
3.1 Application to Black's model: pricing a cap using the forward Libor rate . . . 4
3.2 Pricing of a cap using the forward Libor rate in the general case 5

4 Pricing of a swap **5**

5 Application to the Hull–White model **6**

6 Appendix **6**

7 Bibliography **7**

1 Forward Probability – Change of Numeraire

A numeraire is a unit in which a probability is risk-neutral.

To choose the numeraire $B(t,T)$ is equivalent to investing one's assets in money as of the future date T, since one exactly knows the value of the zero-coupon at time T; this means that we will use as a pricing measure the forward probability $\mathbb{Q}^T$.

On the contrary, if one invests one's assets in the cash numeraire, one invests in today's money (today being date 0), which means that we will use as a pricing measure the risk-neutral probability $\mathbb{Q}$.

Under the risk-neutral probability $\mathbb{Q}$, the price at time t of a zero-coupon bond $B(t,T)$ maturing at time T is solution of the following SDE:

$$\mathrm{d}B(t,T) = B(t,T)\big(r(t)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t\big) \qquad (*)$$

Let F be a random flow, $\mathcal{F}_T$-measurable. Then the price at time t of F is:

$$\pi_t(F) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r_s\,\mathrm{d}s} F \,\middle|\, \mathcal{F}_t\right]$$

We are going to make a change of numeraire and take as the new numeraire the zero-coupon bond maturing at time T.

Under $\mathbb{Q}$, $\dfrac{\pi_t(F)}{S_t^0}$ is a martingale, where $S_t^0 = e^{\int_0^t r_s\,\mathrm{d}s}$ is the money-market numeraire.

We look for a probability $\mathbb{Q}^T$ under which $\pi_t(F)$ is a martingale once expressed in the new numeraire $B(t,T)$.

If $\mathbb{Q}^T$ exists, then $\dfrac{\pi_t(F)}{B(t,T)}$ is a martingale under $\mathbb{Q}^T$ by definition of $\mathbb{Q}^T$. We then obtain, using $B(T,T)=1$:

$$\begin{aligned}\frac{\pi_t(F)}{B(t,T)} &= \mathbb{E}^{\mathbb{Q}^T}\left[\frac{\pi_T(F)}{B(T,T)}\,\middle|\,\mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}^T}[F \mid \mathcal{F}_t]\\ \Rightarrow\quad \pi_t(F) &= B(t,T)\,\mathbb{E}^{\mathbb{Q}^T}[F \mid \mathcal{F}_t]\end{aligned}$$

For $t=0$:

$$\pi_0(F) = B(0,T)\,\mathbb{E}^{\mathbb{Q}^T}[F] \tag{1}$$

By definition, we also have:

$$\pi_0(F) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_s\,\mathrm{d}s}F\right] \tag{2}$$

From (1) and (2), one deduces:

$$B(0,T)\,\mathbb{E}^{\mathbb{Q}^T}[F] = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_s\,\mathrm{d}s}F\right] \quad\Longrightarrow\quad \left.\frac{\mathrm{d}\mathbb{Q}^T}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = B(0,T)^{-1}e^{-\int_0^T r_s\,\mathrm{d}s}$$

One can verify, by applying Itô's formula to the exponential function, that the solution of equation $(*)$ is:

$$B(t,T) = B(0,T)\,\exp\left(\int_0^t r_s\,\mathrm{d}s - \frac{1}{2}\int_0^t \sigma(s,T)^2\,\mathrm{d}s + \int_0^t \sigma(s,T)\,\mathrm{d}W_s\right)$$

For $t=T$, using $B(T,T)=1$, we obtain:

$$1 = B(0,T)\,\exp\left(\int_0^T r_s\,\mathrm{d}s - \frac{1}{2}\int_0^T \sigma(s,T)^2\,\mathrm{d}s + \int_0^T \sigma(s,T)\,\mathrm{d}W_s\right)$$

$$\Rightarrow\quad \left.\frac{\mathrm{d}\mathbb{Q}^T}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = B(0,T)^{-1}e^{-\int_0^T r_s\,\mathrm{d}s} = \exp\left(-\frac{1}{2}\int_0^T \sigma(s,T)^2\,\mathrm{d}s + \int_0^T \sigma(s,T)\,\mathrm{d}W_s\right)$$

The Girsanov theorem then gives us that $\widetilde{W}_t = W_t - \int_0^t \sigma(s,T)\,\mathrm{d}s$ is a Brownian motion under $\mathbb{Q}^T$.

Converse verification. Let us show that if $e^{-\int_0^t r_s\,\mathrm{d}s}\pi_t(F)$ is a martingale under $\mathbb{Q}$, then $\dfrac{\pi_t(F)}{B(t,T)}$ is a martingale under $\mathbb{Q}^T$.

We know that:

$$e^{-\int_0^t r_s\,\mathrm{d}s}\pi_t(F) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_s\,\mathrm{d}s}\pi_T(F) \mid \mathcal{F}_t\right] = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r_s\,\mathrm{d}s}F \mid \mathcal{F}_t\right]$$

For any $B \in \mathcal{F}_t$, using the density $\left.\dfrac{\mathrm{d}\mathbb{Q}^T}{\mathrm{d}\mathbb{Q}}\right|_{\mathcal{F}_T} = \exp\left(-\frac{1}{2}\int_0^T \sigma(s,T)^2\mathrm{d}s + \int_0^T \sigma(s,T)\mathrm{d}W_s\right)$:

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}^T}[F\,\mathbf{1}_{\{B\}}] &= B(0,T)^{-1}\,\mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^t r_s\,\mathrm{d}s}\,\pi_t(F)\,\mathbf{1}_{\{B\}}\right]\\ &= \mathbb{E}^{\mathbb{Q}^T}\left[\frac{\pi_t(F)\,\mathbf{1}_{\{B\}}}{B(t,T)}\right]\end{aligned}$$

Since this holds for every $B \in \mathcal{F}_t$, we obtain:

$$\frac{\pi_t(F)}{B(t,T)} = \mathbb{E}^{\mathbb{Q}^T}[F \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^T}\left[\frac{\pi_T(F)}{B(T,T)} \,\middle|\, \mathcal{F}_t\right]$$

which shows that $\dfrac{\pi_t(F)}{B(t,T)}$ is a martingale under $\mathbb{Q}^T$.

A useful result

If $B(t,T)$ satisfies $\mathrm{d}B(t,T) = B(t,T)\big(r(t)\,\mathrm{d}t + \sigma(t,T)\,\mathrm{d}W_t\big)$, then, writing D for a fixed tenor:

$$\mathrm{d}\left(\frac{B(t,T+D)}{B(t,T)}\right) = \frac{B(t,T+D)}{B(t,T)}\big(\sigma(t,T+D) - \sigma(t,T)\big)\,\mathrm{d}\widetilde{W}_t$$

where $\widetilde{W}_t = W_t - \int_0^t \sigma(s,T)\,\mathrm{d}s$ is a Brownian motion under $\mathbb{Q}^T$.

Sketch. From the explicit solution above,

$$\frac{B(t,T+D)}{B(t,T)} = \frac{B(0,T+D)}{B(0,T)}\exp\left(-\frac{1}{2}\int_0^t \big(\sigma(s,T+D)^2 - \sigma(s,T)^2\big)\mathrm{d}s + \int_0^t \big(\sigma(s,T+D) - \sigma(s,T)\big)\mathrm{d}W_s\right)$$

Writing $\mathrm{d}W_s = \mathrm{d}\widetilde{W}_s + \sigma(s,T)\,\mathrm{d}s$ and completing the square inside the exponential gives:

$$\frac{B(t,T+D)}{B(t,T)} = \frac{B(0,T+D)}{B(0,T)}\exp\left(-\frac{1}{2}\int_0^t \big(\sigma(s,T+D) - \sigma(s,T)\big)^2\mathrm{d}s + \int_0^t \big(\sigma(s,T+D) - \sigma(s,T)\big)\mathrm{d}\widetilde{W}_s\right)$$

Applying Itô's formula to this exponential yields the announced SDE.

1.1 Money market contract

A money market contract is an agreement signed at the current time t to lend an amount of money over $[T, T+\delta]$, taking into account the Libor interest paid over that period. If, at inception, the contract is fairly priced (zero value), then for a notional N:

$$N\,e^{-\int_0^T r_s\,\mathrm{d}s}\Big(B(t,T) - (1+\delta\,\mathrm{Libor})B(t,T+\delta)\Big) = 0$$

Taking the $\mathbb{Q}$-expectation and using $B(0,T) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^T r_s\,\mathrm{d}s}\Big] = \mathbb{E}^{\mathbb{Q}}\Big[B(t,T)\,e^{-\int_0^t r_s\,\mathrm{d}s}\Big]$, one obtains:

$$B(0,T) - B(0,T+\delta) - \delta\,B(0,T+\delta)\,\mathbb{E}^{\mathbb{Q}}[\mathrm{Libor}] = 0$$

$$\Rightarrow \quad \mathbb{E}^{\mathbb{Q}}[\mathrm{Libor}] = \frac{B(0,T) - B(0,T+\delta)}{\delta\,B(0,T+\delta)} = \left(\frac{B(0,T)}{B(0,T+\delta)} - 1\right)\frac{1}{\delta} =: L(0,T,T+\delta) \quad \text{(Forward Libor rate)}$$

2 Pricing of a bond option

2.1 Pricing an option on a zero-coupon bond

This option is exercisable at time T on a zero-coupon bond maturing at time $T+D$. The pay-off is $[B(T,T+D) - K]^+$.

The price C_t of the option at time t is:

$$C_t = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_t^T r_s\,\mathrm{d}s}\big[B(T,T+D) - K\big]^+ \,\Big|\, \mathcal{F}_t\Big] = B(t,T)\,\mathbb{E}^{\mathbb{Q}^T}\left[\left[\frac{B(T,T+D)}{B(T,T)} - K\right]^+ \,\middle|\, \mathcal{F}_t\right]$$

We saw that, under $\mathbb{Q}^T$, $Y_t := \dfrac{B(t,T+D)}{B(t,T)}$ satisfies $\mathrm{d}Y_t = Y_t\big(\sigma(t,T+D)-\sigma(t,T)\big)\mathrm{d}\widetilde{W}_t$, hence is a martingale under $\mathbb{Q}^T$.

1st case: σ deterministic. Then Y_T is log-normal under $\mathbb{Q}^T$ and we obtain Black-type formulas:

$$C_t = B(t,T)\big[Y_t N(d_1) - K N(d_2)\big], \qquad P_t = B(t,T)\big[K N(-d_2) - Y_t N(-d_1)\big]$$

where

$$d_2 = \frac{1}{\Sigma_t}\ln\!\left(\frac{Y_t}{K}\right) - \frac{\Sigma_t}{2}, \qquad d_1 = d_2 + \Sigma_t, \qquad \Sigma_t^2 = \int_t^T \big(\sigma(s,T+D)-\sigma(s,T)\big)^2\,\mathrm{d}s$$

2nd case: σ general. At $t=0$:

$$\begin{aligned}
C_0 &= \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^T r_s\,\mathrm{d}s} B(T,T+D)\,\mathbf{1}_{\{B(T,T+D)>K\}}\Big] - K\,\mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^T r_s\,\mathrm{d}s}\,\mathbf{1}_{\{B(T,T+D)>K\}}\Big] \\
&= \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^T r_s\,\mathrm{d}s}\, e^{-\int_T^{T+D} r_s\,\mathrm{d}s}\,\mathbf{1}_{\{B(T,T+D)>K\}}\Big] - K\,B(0,T)\,\mathbb{E}^{\mathbb{Q}^T}\big[\mathbf{1}_{\{B(T,T+D)>K\}}\big] \\
&= \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^{T+D} r_s\,\mathrm{d}s}\,\mathbf{1}_{\{B(T,T+D)>K\}}\Big] - K\,B(0,T)\,\mathbb{Q}^T\big(B(T,T+D)>K\big) \\
&= B(0,T+D)\,\mathbb{Q}^{T+D}\big(B(T,T+D)>K\big) - K\,B(0,T)\,\mathbb{Q}^T\big(B(T,T+D)>K\big)
\end{aligned}$$

where we used $B(T,T+D) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_T^{T+D} r_s\,\mathrm{d}s} \mid \mathcal{F}_T\Big]$ and the tower property. Hence:

$$C_0 = B(0,T+D)\,N(d_1) - K\,B(0,T)\,N(d_2), \qquad P_0 = K\,B(0,T)\,N(-d_2) - B(0,T+D)\,N(-d_1)$$

2.2 Pricing of a bond option at time 0

Consider a coupon bond maturing at T_D, paying coupons C_i at dates $T_i \in \{T_1,\dots,T_D\}$ with $0 \le t \le T_0 = T \le T_1 \le \cdots \le T_D$:

$$O(t,T_D) = \sum_{i=1}^n C_i\,B(t,T_i)$$

For an option of maturity T and strike K on $O(\cdot,T_D)$, the same reasoning as in §2.1 (applied term by term, using $B(T,T_i) = \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_T^{T_i} r_s\,\mathrm{d}s} \mid \mathcal{F}_T\Big]$) gives, at $t=0$:

$$C_0 = \sum_{i=1}^n C_i\,B(0,T_i)\,\mathbb{Q}^{T_i}\left(\sum_{i=1}^n C_i B(T,T_i) > K\right) - K\,B(0,T)\,\mathbb{Q}^T\left(\sum_{i=1}^n C_i B(T,T_i) > K\right)$$

3 Pricing of a cap

3.1 Application to Black's model: pricing a cap using the forward Libor rate

A cap is a portfolio of caplets, each caplet being an option on a future Libor rate maturing at T_i. The Libor's value at T_i is $L(T_i)$, and its forward value at t is $L(t,T_i)$, satisfying:

$$\frac{B(t,T_{i-1})}{B(t,T_i)} = 1 + L(t,T_i)(T_i - T_{i-1}), \qquad 0 \le t \le T_0 = T \le T_1 \le \cdots \le T_n$$

We assume the forward Libor rate over $[T_{i-1}, T_i]$ follows:

$$\frac{\mathrm{d}L(t,T_i)}{L(t,T_i)} = \sigma_i \,\mathrm{d}W_t^i \quad \Longrightarrow \quad L(T_i) = L(0,T_i)\, e^{\sigma_i W^i_{T_i} - \frac{\sigma_i^2}{2} T_i}$$

where W^i is a Brownian motion under $\mathbb{Q}^{T_i}$.

The pay-off of each caplet is $[L(T_i) - K]^+ \delta$, with $\delta := T_i - T_{i-1}$. The price of the cap at time 0 is:

$$\begin{aligned}
\mathrm{Cap} &= \sum_{i=1}^n \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_0^{T_i} r_s \,\mathrm{d}s}[L(T_i) - K]^+ \delta\Big] \\
&= \sum_{i=1}^n \delta\, B(0,T_i)\, \mathbb{E}^{\mathbb{Q}^{T_i}}\left[\left[L(0,T_i)\, e^{\sigma_i W^i_{T_i} - \frac{\sigma_i^2}{2} T_i} - K\right]^+\right] \\
&= \sum_{i=1}^n \delta\, B(0,T_i)\, \Big[L(0,T_i)\, N(d_1^i) - K\, N(d_2^i)\Big]
\end{aligned}$$

with the usual Black convention $d_2^i = \dfrac{1}{\sigma_i\sqrt{T_i}} \ln\left(\dfrac{L(0,T_i)}{K}\right) - \dfrac{\sigma_i\sqrt{T_i}}{2}$, $d_1^i = d_2^i + \sigma_i\sqrt{T_i}$.

3.2 Pricing of a cap using the forward Libor rate in the general case

Here we only assume $\dfrac{1}{B(T_{i-1},T_i)} = 1 + L(T_i)(T_i - T_{i-1})$. Writing $\delta := T_i - T_{i-1}$ and $\kappa := (1 + K\delta)^{-1}$, the caplet payoff can be rewritten as a claim on $B(T_{i-1},T_i)$ paid at T_{i-1}:

$$\begin{aligned}
\mathrm{Cap}_t &= \sum_{i=1}^n \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_t^{T_i} r_s \,\mathrm{d}s}[L(T_i) - K]^+ \delta \,\Big|\, \mathcal{F}_t\Big] \\
&= \sum_{i=1}^n \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^{T_{i-1}} r_s \,\mathrm{d}s}\, B(T_{i-1},T_i)\left(\frac{1}{B(T_{i-1},T_i)} - \kappa\right)^+ \,\middle|\, \mathcal{F}_t\right] \\
&= \sum_{i=1}^n \delta\, B(t,T_{i-1})\, \mathbb{E}^{\mathbb{Q}^{T_{i-1}}}\Big[\big(\kappa^{-1} - B(T_{i-1},T_i)\big)^+ \,\Big|\, \mathcal{F}_t\Big] \\
&= \sum_{i=1}^n \delta\, B(t,T_{i-1})\, P\big(B(T_{i-1},T_i), \kappa^{-1}\big)
\end{aligned}$$

where $P(B(T_{i-1},T_i), \kappa^{-1})$ denotes the price of a put on the zero-coupon bond $B(T_{i-1},T_i)$ with strike κ^{-1}, computed as in §2.1. At $t = 0$:

$$\mathrm{Cap}_0 = \sum_{i=1}^n \delta\, B(0,T_{i-1})\, \Big[\kappa^{-1} B(0,T_{i-1})\, N(-d_2^i) - B(0,T_i)\, N(-d_1^i)\Big]$$

4 Pricing of a swap

Consider a payer swap paying fixed rate k and receiving floating Libor $L(T_i)$ on notional N, with $T_0 = T \le T_1 \le \cdots \le T_n$:

$$\mathrm{PSwap}_t(k) = N \sum_{i=1}^n \mathbb{E}^{\mathbb{Q}}\Big[e^{-\int_t^{T_i} r_s \,\mathrm{d}s}\big(L(T_i) - k\big)\delta \,\Big|\, \mathcal{F}_t\Big]$$

Changing numeraire to $B(\cdot,T_i)$ for each term and using $\mathbb{E}^{\mathbb{Q}^{T_i}}[L(T_i)\mid\mathcal{F}_t]=L(t,T_i)$:

$$\mathrm{PSwap}_t(k)=N\sum_{i=1}^{n}\delta\,B(t,T_i)\big(L(t,T_i)-k\big)$$

Using $\delta\,B(t,T_i)L(t,T_i)=B(t,T_{i-1})-B(t,T_i)$ and telescoping the sum:

$$\mathrm{PSwap}_t(k)=N\left(B(t,T_0)-\sum_{i=1}^{n}c_i\,B(t,T_i)\right),\qquad c_i:=k\delta\ (i<n),\quad c_n:=1+k\delta$$

The forward swap rate $k(t,T,n)$ is the value of k for which $\mathrm{PSwap}_t(k)=0$:

$$k(t,T,n)=\frac{B(t,T_0)-B(t,T_n)}{\delta\sum_{i=1}^{n}B(t,T_i)}$$

5 Application to the Hull–White model

The Hull–White model assumes that the short rate $r(t)$ follows the mean-reverting process:

$$\mathrm{d}r(t)=\big(\theta(t)-a\,r(t)\big)\,\mathrm{d}t+\sigma(t)\,\mathrm{d}W_t$$

The price at t of a discount bond maturing at T is:

$$B(t,T)=A(t,T)\,e^{-P(t,T)r(t)},\qquad P(t,T):=\frac{1-e^{-a(T-t)}}{a}$$

$$A(t,T)=\exp\!\left(-\int_t^T\theta(u)P(t,u)\,\mathrm{d}u+\frac{1}{2}\int_t^T\sigma(u)^2P(t,u)^2\,\mathrm{d}u\right)$$

We verify that, under $\mathbb{Q}$:

$$\frac{\mathrm{d}B(t,T)}{B(t,T)}=r(t)\,\mathrm{d}t-\sigma(t)\,P(t,T)\,\mathrm{d}W_t$$

Sketch. Writing $\ln B(t,T)=\ln A(t,T)-P(t,T)\,r(t)$ and applying Itô's formula, using $\mathrm{d}\langle B(\cdot,T),B(\cdot,T)\rangle_t=B(t,T)^2P(t,T)^2\sigma(t)^2\,\mathrm{d}t$, one gets:

$$\frac{\mathrm{d}B(t,T)}{B(t,T)}-\frac{P(t,T)^2\sigma(t)^2}{2}\mathrm{d}t=\theta(t)P(t,T)\,\mathrm{d}t+\frac{1}{2}\sigma(t)^2P(t,T)^2\,\mathrm{d}t-\mathrm{d}P(t,T)\,r(t)-P(t,T)\,\mathrm{d}r(t)$$

The definitions of $A(t,T)$ and $P(t,T)$ are precisely chosen so that the drift terms collapse to $r(t)\,\mathrm{d}t$, yielding the announced SDE with volatility $-\sigma(t)P(t,T)$.

6 Appendix

Theorem 1 (Girsanov). *If W_t is a Brownian motion under a probability P and γ_t is an $\mathcal{F}_t$-previsible process such that $\mathbb{E}^P\Big[\exp\Big(\frac{1}{2}\int_0^T\gamma_t^2\,\mathrm{d}t\Big)\Big]<\infty$, then there exists a probability Q such that:*

1. *Q is equivalent to P;*
2. *$\dfrac{\mathrm{d}Q}{\mathrm{d}P}=L_T=\exp\Big(-\int_0^T\gamma_t\,\mathrm{d}W_t-\frac{1}{2}\int_0^T\gamma_t^2\,\mathrm{d}t\Big)$;*
3. *$\widetilde{W}_t=W_t+\int_0^t\gamma_s\,\mathrm{d}s$ is a Brownian motion under Q.*

Proof. Intuitively, since $W_T \sim \mathcal{N}(0,T)$ under P:

$$\mathbb{E}^P\left[e^{\theta W_T}\right] = e^{\theta^2 T/2}$$

$$\mathbb{E}^Q\left[e^{\theta W_T}\right] = \mathbb{E}^P\left[e^{-\gamma W_T - \gamma^2 T/2}\, e^{\theta W_T}\right] = \mathbb{E}^P\left[e^{-(\gamma-\theta)W_T - \gamma^2 T/2}\right] = e^{-\gamma\theta T + \theta^2 T/2}$$

which shows that W_T, under Q, has the law $\mathcal{N}(-\gamma T, T)$ – consistent with $\widetilde{W}_T = W_T + \gamma T$ being $\mathcal{N}(0,T)$.

To establish this rigorously for $\gamma_s \equiv \gamma$ constant, we first check that $L_t = \exp(-\gamma W_t - \gamma^2 t/2)$ is a P-martingale:

$$\mathbb{E}^P[L_t \mid \mathcal{F}_s] = \mathbb{E}^P\left[e^{-\gamma(W_t - W_s)}\right] e^{-\gamma W_s - \gamma^2 t/2} = e^{-\gamma^2(t-s)/2}\, e^{-\gamma W_s - \gamma^2 t/2} = e^{-\gamma W_s - \gamma^2 s/2} = L_s$$

For any $\mathcal{F}_t$-martingale L_t (with $L_0 = 1$) defining Q via $\mathrm{d}Q/\mathrm{d}P|_{\mathcal{F}_T} = L_T$, and any $\mathcal{F}_t$-measurable Z, one has the standard Bayes formula $\mathbb{E}^Q[Z \mid \mathcal{F}_t] = L_t^{-1}\mathbb{E}^P[ZL_T \mid \mathcal{F}_t]$. Applying it with $Z = e^{iu(W_t - W_s)}$:

$$\mathbb{E}^Q\left[e^{iu(W_t - W_s)} \mid \mathcal{F}_s\right] = \frac{\mathbb{E}^P\left[e^{iu(W_t - W_s)}\, L_t \mid \mathcal{F}_s\right]}{L_s} = e^{-u^2(t-s)/2}$$

which is the characteristic function of $\mathcal{N}(0, t-s)$, independent of $\mathcal{F}_s$. Hence $\widetilde{W}_t = W_t + \gamma t$ has independent, stationary $\mathcal{N}(0, t-s)$ increments under Q: it is a Brownian motion under Q. □

7 Bibliography

References

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